

UNIT-III Electromagnetic Waves

Stationary charge $\rightarrow$ Electrostatic field
 Moving charge (steady current) $\rightarrow$ Magnetostatic field
 Time varying current $\rightarrow$ Electromagnetic field

Maxwell's Equation

$\nabla \cdot \vec{B} = 0$ $\nabla \cdot \vec{D} = \rho$ $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$ $\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$	$\oint \vec{B} \cdot d\vec{s} = 0$ $\oint \vec{D} \cdot d\vec{s} = \int \rho \cdot dV$ $\oint \vec{E} \cdot d\vec{l} = -\int \frac{\partial \vec{B}}{\partial t} \cdot d\vec{s}$ $\oint \vec{H} \cdot d\vec{l} = \int (\vec{J} + \frac{\partial \vec{D}}{\partial t}) \cdot d\vec{s}$	$\rightarrow$ Gauss law for static magnetic field (There are no magnetic monopoles) $\rightarrow$ Gauss ^{law} for Electric field $\rightarrow$ A time varying magnetic field can produce an electric field $\rightarrow$ A time varying $\vec{E}$ and an $\vec{E}$ produces a magnetic field.
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where D = electric displacement (flux) density (C/m^2)

B = Magnetic flux density (W/m^2)

E = Electric field Intensity (V/m)

H = magnetic field intensity (A/m)

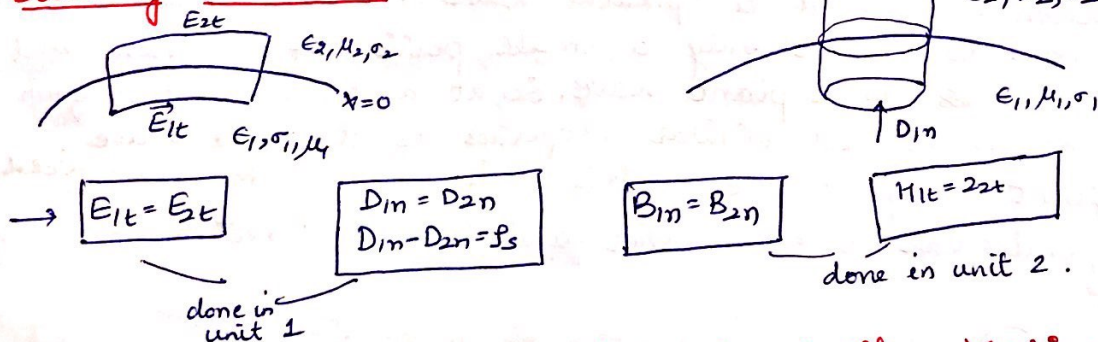
J = conduction current density (A/m^2)

ρ = charge density (C/m^3)

$$\begin{aligned}\vec{D} &= \epsilon \vec{E} \\ \vec{B} &= \mu \vec{H} \\ \vec{J} &= \sigma \vec{E}\end{aligned}$$

The eqⁿ $\nabla \times \vec{E}$ and $\nabla \times \vec{H}$ implies that time varying electric and magnetic fields in empty space are independent re. a changing $\vec{E}$ can produce a magnetic field and vice versa, which implies that a time changing electromagnetic field would propagate energy through empty space with the speed of light.

Boundary Conditions



Wave Equation for Free Space / Non-Conducting Medium

Free space / Non-conducting medium $\rightarrow J=0$ & $\rho=0$

$$\therefore \nabla \cdot \vec{B} = 0 \quad \text{--- (A)}$$

$$\nabla \cdot \vec{D} = 0 \quad \text{--- (B)}$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \text{--- (C)}$$

$$\text{and } \nabla \times \vec{H} = \frac{\partial \vec{D}}{\partial t} \quad \text{--- (D)}$$

Differentiating (D)

$$\frac{\partial}{\partial t} (\nabla \times \vec{H}) = \frac{\partial^2 \vec{D}}{\partial t^2}$$

$$\nabla \times \left(\frac{\partial \vec{H}}{\partial t} \right) = \epsilon \frac{\partial^2 \vec{E}}{\partial t^2}$$

Since ∇ = differentiation wrt space, order can be changed

Taking curl on eq (3)/c

$$\nabla \times \nabla \times \vec{E} = -\nabla \times \frac{\partial \vec{B}}{\partial t} = -\nabla \times \frac{\partial \mu_0 \vec{H}}{\partial t} = -\mu_0 \nabla \times \frac{\partial \vec{H}}{\partial t} = -\mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\boxed{\nabla \times \nabla \times \vec{E} = -\mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}}$$

Using vector identity $\vec{A} \times \vec{B} \times \vec{C} = (\vec{A} \cdot \vec{C})\vec{B} - (\vec{A} \cdot \vec{B})\vec{C}$
 $\nabla \times \nabla \times \vec{E} = (\nabla \cdot \vec{E})\nabla - (\nabla \cdot \nabla)\vec{E} = (\nabla \cdot \vec{E})\nabla - \nabla^2 \vec{E}$

$$(\nabla \cdot \vec{E})\nabla - \nabla^2 \vec{E} = -\mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\begin{aligned} \nabla \cdot \vec{D} &= 0 \\ \nabla \cdot \epsilon \vec{E} &= 0 \\ \nabla \cdot \vec{E} &= 0 \end{aligned}$$

$$\therefore -\nabla^2 \vec{E} = -\mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\boxed{\nabla^2 \vec{E} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}} \text{ - Wave Equation in term of electric field intensity}$$

Similarly $\boxed{\nabla^2 \vec{H} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{H}}{\partial t^2}} \rightarrow \text{Wave Equation in term of magnetic field intensity}$

The standard partial differential wave eq frequently encountered in mechanical engineering is $\nabla^2 X = \frac{1}{v^2} \frac{\partial^2 X}{\partial t^2}$
 $X \rightarrow$ desired field vector/component
 $v \rightarrow$ velocity of wave

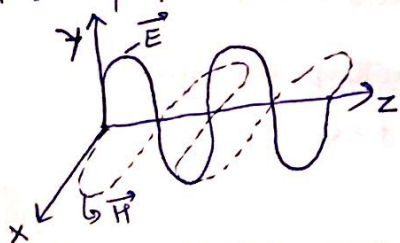
Comparing $v^2 = \frac{1}{\mu_0 \epsilon_0} = \frac{1}{4\pi \times 10^{-7} \times \frac{1}{36\pi \times 10^9}} = 3 \times 10^8 \text{ m/s}$

So, an EM wave travels with velocity of light (v) here denoted by c .

Properties of EM waves in Free Space

An EM wave originates from a point in free space, spreads out uniformly in all direction & it forms a spherical wave. An observer, at a great distance is able to observe only a small portion of the wave and hence it appears as a plane wave. So, at a large distance from the source the wave has similar properties to the plane wave.

1) At each point in space, the electric vector $\vec{E}$ and magnetic field $\vec{H}$ are perpendicular to each other and to the direction of propagation.



$Z =$ direction of propagation
 $\vec{E} =$ along Y-axis
 $\vec{H} =$ along X-axis

In a general EM wave, $\vec{E}$ lies in Y or Z plane and $\vec{H}$ in X or Z plane. Since $\vec{E}$ and $\vec{H}$ are transverse (90°) to the direction of propagation. So, they are called Transverse Electromagnetic (TEM) wave.

② Velocity of propagation of wave in free space is given by $c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$
 whatever may be the frequency, EM wave travels in space with velocity of light.

constant being equal to 120π or 377 or $\sqrt{\frac{\mu_0}{\epsilon_0}}$

$$\frac{E}{H} = 120\pi = 377 = \sqrt{\frac{\mu_0}{\epsilon_0}}$$

Solution of Maxwell Equation in Free Space

Consider an EM wave that is travelling in z -direction then $\frac{\partial E}{\partial x} = 0$ and $\frac{\partial E}{\partial y} = 0$ since it is propagating in z -direction it will have E_x and E_y but not E_z i.e. $E_z = 0$.

$$\nabla^2 E = \mu_0 \epsilon_0 \frac{\partial^2 E}{\partial t^2}$$

$$\left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right] E_x \hat{a}_x + \left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right] E_y \hat{a}_y + \left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right] E_z \hat{a}_z = \mu_0 \epsilon_0 \frac{\partial^2}{\partial t^2} [E_x \hat{a}_x + E_y \hat{a}_y + E_z \hat{a}_z]$$

$$\frac{\partial E_x}{\partial x} = 0 \Rightarrow \frac{\partial^2 E_x}{\partial x^2} = 0$$

$$\frac{\partial E_y}{\partial y} = 0 \Rightarrow \frac{\partial^2 E_y}{\partial y^2} = 0$$

$$\& E_z = 0$$

$$\frac{\partial^2 E_x}{\partial z^2} \hat{a}_x + \frac{\partial^2 E_y}{\partial z^2} \hat{a}_y = \mu_0 \epsilon_0 \frac{\partial^2}{\partial t^2} [E_x \hat{a}_x + E_y \hat{a}_y]$$

$$\text{or } \boxed{\begin{aligned} \frac{\partial^2 E_x}{\partial z^2} &= \mu_0 \epsilon_0 \frac{\partial^2 E_x}{\partial t^2} \\ \frac{\partial^2 E_y}{\partial z^2} &= \mu_0 \epsilon_0 \frac{\partial^2 E_y}{\partial t^2} \end{aligned}}$$

$$\text{or } \boxed{\begin{aligned} \frac{\partial^2 H_x}{\partial z^2} &= \mu_0 \epsilon_0 \frac{\partial^2 H_x}{\partial t^2} \\ \frac{\partial^2 H_y}{\partial z^2} &= \mu_0 \epsilon_0 \frac{\partial^2 H_y}{\partial t^2} \end{aligned}} \quad \text{--- (1)}$$

The above equation relates space and time variation of the scalar magnitude of E_x or E_y

The above equations are 2nd order partial differential equation & the general solution of such differential equation are -

$$E = f_1(z-ct) + f_2(z+ct)$$

wave travelling in positive z -direction

negative z -direction.

When there is no reflecting surface so that the reflected wave travels back to the source, the second term would be zero i.e.

$$E = f_1(z-ct)$$

The general solution of the eq (1) can be put in trigonometric form as

$$E_y = \sin \beta(z+mt) \quad \text{--- (2)}$$

$$\boxed{\beta = \frac{2\pi}{\lambda}, m = \text{constant}}$$

Put eq (2) in (1)

Apply 2nd derivation, and we have the solution as $m = \pm c$

Hence $E_y = \sin \beta(z-ct) + \sin \beta(z+ct)$ is the general solution.

$$\because \beta = \frac{2\pi}{\lambda}$$

$$\beta c = \frac{2\pi c}{\lambda} = \frac{2\pi}{\lambda} \cdot f \lambda = 2\pi f = \omega$$

If only +ve direction wave is taken

$$E_y = \sin(\beta z - \omega t)$$

$\therefore E_y = \sin(\omega t - \beta z) \rightarrow$ It has a difference of phase of π radian.

If the phase displacement is disregarded, then

$$E_y = E_0 \sin(\omega t - \beta z) \leftarrow \text{when maximum amplitude is not 1 but } E_0$$

The solution in exponential form can also be given as

$$E_y = E_0 e^{j(\omega t + \beta z)}$$

$$E_y = E_0 \sin \omega \left(t - \frac{\beta z}{\omega} \right) = E_0 \sin \omega \left(t - \frac{2\pi \cdot z}{2\pi f} \right) = E_0 \sin \omega \left(t - \frac{z}{v} \right)$$

$$\begin{aligned} E_y &= E_0 \sin \omega \left(t - \frac{z}{v} \right) \\ E_y &= E_0 \sin \omega (t - t_p) \\ E_y &= E_0 \sin(\omega t - \beta z) \end{aligned}$$

where $t_p = \frac{z}{v}$ = time of propagation of wave from origin of wave to a point P of observation

Transverse Nature of Uniform Plane Wave [ET2017, 2019, 6M]

A Maxwell wave equation in $\vec{E}$ is given by

$$\nabla^2 \vec{E} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}$$

Let the wave be propagating in z-direction, so for a uniform plane wave $\frac{\partial}{\partial x} = \frac{\partial}{\partial y} = 0 \Rightarrow \frac{\partial^2}{\partial x^2} = 0 = \frac{\partial^2}{\partial y^2} = 0$

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) E_x \hat{a}_x + \left(\frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial z^2} \right) E_y \hat{a}_y + \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) E_z \hat{a}_z = \mu_0 \epsilon_0 \frac{\partial^2}{\partial t^2} [E_x \hat{a}_x + E_y \hat{a}_y + E_z \hat{a}_z]$$

$$\therefore \frac{\partial^2 E_x}{\partial z^2} \hat{a}_x + \frac{\partial^2 E_y}{\partial z^2} \hat{a}_y + \frac{\partial^2 E_z}{\partial z^2} \hat{a}_z = \mu_0 \epsilon_0 \frac{\partial^2}{\partial t^2} [E_x \hat{a}_x + E_y \hat{a}_y + E_z \hat{a}_z]$$

$$\therefore \frac{\partial^2 E_x}{\partial z^2} \hat{a}_x + \frac{\partial^2 E_y}{\partial z^2} \hat{a}_y + \frac{\partial^2 E_z}{\partial z^2} \hat{a}_z = \mu_0 \epsilon_0 \frac{\partial^2}{\partial t^2} [E_x \hat{a}_x + E_y \hat{a}_y + E_z \hat{a}_z]$$

$$\therefore \frac{\partial^2 E_x}{\partial z^2} = \mu_0 \epsilon_0 \frac{\partial^2 E_x}{\partial t^2}, \quad \frac{\partial^2 E_y}{\partial z^2} = \mu_0 \epsilon_0 \frac{\partial^2 E_y}{\partial t^2}, \quad \frac{\partial^2 E_z}{\partial z^2} = \mu_0 \epsilon_0 \frac{\partial^2 E_z}{\partial t^2} \quad \text{--- (1)}$$

$$\text{In free space } \rho = 0 \Rightarrow \begin{aligned} \nabla \cdot \vec{D} &= 0 \\ \nabla \cdot \vec{E} &= 0 \\ \nabla \times \vec{E} &= 0 \end{aligned}$$

$$\frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} = 0$$

For a uniform plane wave E is not dependent on the x or y component

$$\text{i.e. } \frac{\partial E_x}{\partial x} = 0 \text{ and } \frac{\partial E_y}{\partial y} = 0 \quad \therefore \frac{\partial E_z}{\partial z} = 0 \Rightarrow \frac{\partial^2 E_z}{\partial z^2} = 0 \quad \text{--- (2)}$$

which implies that there is no variation of component E_z in z-direction.

$$\text{Put (2) in (1)} \quad \mu_0 \epsilon_0 \frac{\partial^2 E_z}{\partial t^2} = 0$$

which implies E_z must be $\left\{ \begin{array}{l} \text{zero or} \\ \text{constant in time or} \\ \text{increasing uniformly with time} \end{array} \right\}$ are not properties of wave

$$\boxed{\text{So } \Rightarrow E_z = 0}$$

The wave has no z components of $\vec{E}$ and $\vec{H}$ when propagating in z-direction. $\vec{E} \& \vec{H} \perp$ direction of propagation $\Rightarrow$ Transverse wave.

It has been shown that E & H are $\perp$ to direction of propagation.
Now to show that $E \perp H \perp$ direction of propagation.

Acc. to Maxwell's equation

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x & E_y & E_z \end{vmatrix}$$

For uniform plane wave in z -direction
 $\rightarrow E_z = 0, \partial/\partial x = \partial/\partial y = 0$

$$\hat{a}_x \left[0 - \frac{\partial E_y}{\partial z} \right] - \hat{a}_y \left[0 - \frac{\partial E_x}{\partial z} \right] + \hat{a}_z [0 - 0] = -\frac{\partial}{\partial t} [\vec{B}_x \hat{a}_x + \vec{B}_y \hat{a}_y + \vec{B}_z \hat{a}_z]$$

$$\frac{\partial}{\partial z} [E_y \hat{a}_x - E_x \hat{a}_y] = -\hat{a}_x \frac{\partial B_x}{\partial t} - \hat{a}_y \frac{\partial B_y}{\partial t}$$

$$\text{or } \frac{\partial E_y}{\partial z} = \frac{\partial B_x}{\partial t}$$

$$\frac{\partial E_x}{\partial z} = \frac{\partial B_y}{\partial t}$$

$$\text{or } \begin{cases} \frac{\partial E_y}{\partial z} = -\mu \frac{\partial H_x}{\partial t} \\ \frac{\partial E_x}{\partial z} = \mu \frac{\partial H_y}{\partial t} \end{cases} \quad \star$$

From Maxwell's equation

$$\nabla \times \vec{H} = \frac{\partial \vec{D}}{\partial t}$$

By

$$\begin{cases} \frac{\partial H_y}{\partial z} = \epsilon \frac{\partial E_x}{\partial t} \text{ and } \frac{\partial H_x}{\partial z} = -\epsilon \frac{\partial E_y}{\partial t} \end{cases} \quad \star$$

Let the eqn of plane wave be

$$E_y = f_1(z-ct) = F \text{ where } c = \frac{1}{\sqrt{\mu\epsilon}}$$

Diff w.r.t time $\frac{\partial E_y}{\partial t} = f_1'(z-ct) (-c)$

$$\boxed{\frac{\partial E_y}{\partial t} = -c F'} \quad \text{where } F' \text{ denotes first partial differentiation}$$

$$\text{From } \star \quad -\frac{\partial H_x}{\partial z} = \epsilon \frac{\partial E_y}{\partial t} \quad \text{or } -\frac{\partial H_x}{\partial z} = \epsilon (-c F')$$

$$\frac{\partial H_x}{\partial z} = \epsilon c F'$$

Integrating $H_x = \epsilon c F + (K)$ constant of integration but can be neglected as static field is of no interest in radiation process

$$\therefore H_x = \epsilon \cdot \frac{1}{\sqrt{\mu_0 \epsilon_0}} E_y$$

$$\boxed{H_x = \sqrt{\frac{\epsilon_0}{\mu_0}} E_y}$$

$$\text{or } \boxed{\frac{E_y}{H_x} = \sqrt{\frac{\mu}{\epsilon}}}$$

$$\text{By } \frac{E_x}{H_y} = -\sqrt{\frac{\mu}{\epsilon}}$$

$$E = |\vec{E}| = \sqrt{E_x^2 + E_y^2 + 0} = \sqrt{E_x^2 + E_y^2} = \sqrt{\left(\frac{\mu}{\epsilon}\right) H_y^2 + \frac{\mu}{\epsilon} H_x^2} = \sqrt{\frac{\mu}{\epsilon}} \cdot \sqrt{H_x^2 + H_y^2} = \sqrt{\frac{\mu}{\epsilon}} H$$

$$H = |\vec{H}| = \sqrt{H_x^2 + H_y^2 + 0} = \sqrt{H_x^2 + H_y^2}$$

$$\frac{E}{H} = \sqrt{\frac{\mu}{\epsilon}} = \eta$$

$$\eta = \frac{E}{H} = \sqrt{\frac{\mu}{\epsilon}} = \frac{\text{Volt/metre}}{\text{Ampere/metre}} = \frac{V}{A} = \text{Impedance}$$

It is known as characteristic impedance or intrinsic impedance

$$\eta = \eta_0 = \sqrt{\frac{\mu_0}{\epsilon_0}} = \sqrt{\frac{4\pi \times 10^{-7}}{\frac{1}{36\pi \times 10^9}}} = \sqrt{4\pi^2 \times 36 \times 100} = 2\pi \times 6 \times 10 = 120\pi$$

$$\boxed{\eta = 120\pi = 377\Omega}$$

$$\vec{E} \cdot \vec{H} = (\hat{a}_x E_x + \hat{a}_y E_y) \cdot (\hat{a}_x H_x + \hat{a}_y H_y) = E_x H_x + E_y H_y \\ = -\eta H_y H_x + \eta H_x H_y \\ = 0$$

$$\Rightarrow \vec{E} \cdot \vec{H} = 0$$

$$E H \cos \theta = 0 \Rightarrow \theta = 90^\circ \because E \neq 0 \text{ \& } H \neq 0$$

Hence $\vec{E}$ and H are $\perp$ to each other.

Electromagnetic wave Equation for Conducting Medium

Maxwell equation for conducting medium

$$\nabla \cdot \vec{B} = 0$$

$$\nabla \cdot \vec{D} = \rho$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

Since the medium has a conductivity σ , the conduction current density is $\vec{J} = \sigma \vec{E}$

$$\nabla \times \vec{H} = \sigma \vec{E} + \frac{\partial (\epsilon \vec{E})}{\partial t} = \sigma \vec{E} + \epsilon \frac{\partial \vec{E}}{\partial t} \quad \text{--- (1)}$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} = -\mu \frac{\partial \vec{H}}{\partial t}$$

$$\text{Taking curl} \quad \nabla \times \nabla \times \vec{E} = \nabla \times \left(-\mu \frac{\partial \vec{H}}{\partial t} \right) = -\mu \nabla \times \frac{\partial \vec{H}}{\partial t} = -\mu \left(\nabla \times \frac{\partial \vec{H}}{\partial t} \right) \quad \text{--- (2)}$$

$$\text{From (1)} \quad \frac{\partial}{\partial t} (\nabla \times \vec{H}) = \sigma \frac{\partial \vec{E}}{\partial t} + \epsilon \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\nabla \times \frac{\partial \vec{H}}{\partial t} = \sigma \frac{\partial \vec{E}}{\partial t} + \epsilon \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\text{Putting in (2) after using vector identity} \\ \nabla (\nabla \cdot \vec{E}) - \nabla^2 \vec{E} = -\mu \sigma \frac{\partial \vec{E}}{\partial t} - \mu \epsilon \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\begin{aligned} \nabla \cdot \vec{D} &= \rho \\ \nabla \cdot (\epsilon \vec{E}) &= \rho \\ \nabla \cdot \vec{E} &= \rho / \epsilon \end{aligned}$$

$$\nabla^2 \vec{E} = \nabla \left(\frac{\rho}{\epsilon} \right) + \mu \sigma \frac{\partial \vec{E}}{\partial t} + \mu \epsilon \frac{\partial^2 \vec{E}}{\partial t^2}$$

Since there is no net charge within a conductor although it may be on the surface, the charge density $\rho = 0$ hence the first term becomes zero

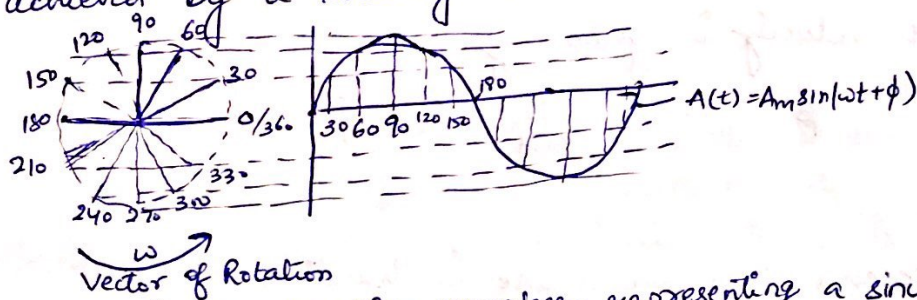
$$\nabla^2 \vec{E} = \mu\sigma \frac{\partial \vec{E}}{\partial t} + \mu\epsilon \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\nabla^2 \vec{H} = \mu\sigma \frac{\partial \vec{H}}{\partial t} + \mu\epsilon \frac{\partial^2 \vec{H}}{\partial t^2}$$

Phasor Notation

Sinusoidal waveform of the same frequency can have a phase difference b/w themselves which represents the angular difference of the two sinusoidal waveform. The term 'lead' and 'lag' as well as 'in phase' and 'out of phase' are commonly used to indicate the relation b/w different waveform with the generalized expression $A(t) = A_m \sin(\omega t \pm \phi)$ representing in the time-domain form.

Such mathematical presentation is difficult to visualize the angular or phasor difference. This is overcome by using phasor diagram, which is achieved by a rotating vector method.



Both vector and phasor are used to describe line that itself it magnitude & direction.
vector $\rightarrow$ peak value
Phase $\rightarrow$ rms value

A phasor is a complex number representing a sinusoidal function whose amplitude, angular frequency and initial phase are time variant. It is a process in which when a sinusoidal function is decomposed into the product of a complex constant and a factor that encapsulates the frequency and time dependence. The complex constant, which encapsulates amplitude and phase dependence is known as phasor. An additional feature of the phasor transform is that differentiation and integration of sinusoidal signal corresponds to simple algebraic operations on the phasor.

Propagation of Wave in Lossless Medium

The wave equation can be written in phasor form as

$$\nabla^2 \vec{E} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\nabla^2 \vec{E} = \mu_0 \epsilon_0 (j\omega)(j\omega) \vec{E}$$

$$\nabla^2 \vec{E} = -\mu_0 \epsilon_0 \omega^2 \vec{E}$$

$$\text{or } \nabla^2 \vec{E} + \mu_0 \epsilon_0 \omega^2 \vec{E} = 0$$

$$= \nabla^2 \vec{E} + \beta^2 \vec{E} = 0 \rightarrow \text{Helmholtz eqn}$$

$$\text{where } \beta^2 = \mu_0 \epsilon_0 \omega^2 \text{ or } \beta = \omega \sqrt{\mu_0 \epsilon_0}$$

-X

$$\therefore \vec{E} = \vec{E}_0 e^{j\omega t}$$

$$\frac{\partial \vec{E}}{\partial t} = \vec{E}_0 (j\omega) e^{j\omega t}$$

$$\frac{\partial \vec{E}}{\partial t} = (j\omega) \vec{E} \Rightarrow \frac{\partial}{\partial t} = j\omega$$

For a plane uniform plane wave propagating in +z direction there are no x & y component. Thus the eqn are

$$\frac{\partial^2 E_x}{\partial z^2} = -\beta^2 E_x \quad \text{and} \quad \frac{\partial^2 E_y}{\partial z^2} = -\beta^2 E_y$$

Ans B. The solution for $\vec{E}_y$ can be written as $\vec{E}_y = A_1 e^{-j\beta z} + A_2 e^{+j\beta z}$ where A_1 and A_2 are arbitrary complex numbers

This is the solution in phasor quantity

$$\begin{aligned} \tilde{E}_y(z, t) &= \text{Real} [E_y(z) e^{j\omega t}] \\ &= \text{Real} [A_1 e^{-j\beta z} + A_2 e^{+j\beta z}) e^{j\omega t}] \\ &= \text{Real} \{ A_1 e^{j(\omega t - \beta z)} + A_2 e^{j(\omega t + \beta z)} \} \end{aligned}$$

$$E_y(z, t) = A_1 \cos(\omega t - \beta z) + A_2 \cos(\omega t + \beta z) \quad \leftarrow \text{This is the solution in time varying field.}$$

From eq(*) the wave velocity is given by

$$\frac{\beta}{\omega} = \sqrt{\epsilon_0 \mu_0} \quad \text{or} \quad \frac{\omega}{\beta} = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = \frac{2\pi f}{2\pi H} = hf = c$$

$$\boxed{c = \frac{\omega}{\beta}}$$

The equation represents two waves, one is travelling in +z direction and the second in negative z-direction. If $A_1 = A_2$ the two waves combine and form a simple standing wave which stands & does not progress.



It is a wave which oscillates in time but whose peak amplitude profile does not move in space. The most common cause of standing wave is resonance.

Propagation of Plane Wave in Conducting Medium

Wave Eqⁿ for conducting medium: $\nabla^2 \vec{E} = \mu\sigma \frac{\partial \vec{E}}{\partial t} + \mu\epsilon \frac{\partial^2 \vec{E}}{\partial t^2}$

By applying phasor notation

$$\nabla^2 \vec{E} = \mu\sigma(j\omega) \vec{E} + \mu\epsilon(j\omega)^2 \vec{E}$$

$$= [\mu j\omega(\sigma + j\omega\epsilon)] \vec{E}$$

$$\nabla^2 \vec{E} = \gamma^2 \vec{E}$$

where $\gamma^2 = \mu j\omega(\sigma + j\omega\epsilon)$

$$\gamma^2 = j\mu\omega\sigma + j^2\mu\omega^2\epsilon$$

$$\gamma^2 = j\mu\omega\sigma - \omega^2\mu\epsilon = \alpha^2 - \beta^2 + j2\alpha\beta$$

comparing

$$\alpha^2 - \beta^2 = -\omega^2\mu\epsilon \text{ and } 2\alpha\beta = \frac{\mu\omega\sigma}{2}$$

Putting value of β

$$-\omega^2\mu\epsilon = \alpha^2 - \frac{\omega^2\mu\sigma^2}{4\alpha^2}$$

$$\alpha^2 + \omega^2\mu\epsilon = \frac{\omega^2\mu\sigma^2}{4\alpha^2}$$

$$\alpha^4 + \alpha^2\omega^2\mu\epsilon = \left(\frac{\omega\mu\sigma}{2}\right)^2$$

Adding $\left[\frac{\omega^2\mu\epsilon}{2}\right]^2$ to both sides

$$\alpha^4 + \omega^2\mu\epsilon \cdot \alpha^2 + \left(\frac{\omega^2\mu\epsilon}{2}\right)^2 = \left(\frac{\omega^2\mu\epsilon}{2}\right)^2 + \left(\frac{\omega\mu\sigma}{2}\right)^2$$

$$\left[\alpha + \frac{\omega^2\mu\epsilon}{2}\right]^2 = \left(\frac{\omega^2\mu\epsilon}{2}\right)^2 \left[1 + \frac{\mu^2\omega^2\sigma^2}{4\omega^2\mu^2\epsilon^2}\right]$$

$$\left[\alpha + \frac{\omega^2\mu\epsilon}{2}\right]^2 = \frac{\omega^4\epsilon^2\mu^2}{4} \left[1 + \frac{\sigma^2}{\omega^2\epsilon^2}\right]$$

Taking square root

$$\alpha + \frac{\omega^2\mu\epsilon}{2} = \frac{\omega^2\epsilon\mu}{2} \sqrt{1 + \left(\frac{\sigma}{\omega\epsilon}\right)^2}$$

$$\alpha^2 = \frac{\omega^2\mu\epsilon}{2} \sqrt{1 + \left(\frac{\sigma}{\omega\epsilon}\right)^2} - \frac{\omega^2\mu\epsilon}{2}$$

$$\alpha^2 = \frac{\omega^2\mu\epsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega\epsilon}\right)^2} - 1 \right]$$

$$\alpha = \pm \sqrt{\frac{\omega^2\mu\epsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega\epsilon}\right)^2} - 1 \right]}$$

$$\alpha = \omega \sqrt{\frac{\mu\epsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega\epsilon}\right)^2} - 1 \right]}$$

$\gamma = \pm \sqrt{j\omega\mu(\sigma + j\omega\epsilon)} = \alpha + j\beta$ phase constant

$\gamma^2 = \alpha^2 - \beta^2 + j2\alpha\beta$ $\rightarrow$ attenuation constant

or $\alpha = \frac{\mu\omega\sigma}{2\beta}$ or $\beta = \frac{\omega\mu\sigma}{2\alpha}$

Putting value of α

$$-\omega^2\mu\epsilon = \alpha^2 - \beta^2 \text{ or } \beta^2 - \omega^2\mu\epsilon = \alpha^2$$

$$\beta^2 - \omega^2\mu\epsilon = \frac{\mu^2\omega^2\sigma^2}{4\beta^2}$$

$$\beta^4 - \omega^2\mu\epsilon \cdot \beta^2 = \left(\frac{\mu\omega\sigma}{2}\right)^2$$

In order to complete square add term

$$\left[-\frac{\omega^2\mu\epsilon}{2}\right]^2 \text{ to both sides}$$

$$\beta^4 - \omega^2\mu\epsilon \cdot \beta^2 + \left[-\frac{\omega^2\mu\epsilon}{2}\right]^2 = \frac{\mu^2\omega^2\sigma^2}{4} + \left[-\frac{\omega^2\mu\epsilon}{2}\right]^2$$

$$\Rightarrow \left[\beta^2 - \frac{\omega^2\mu\epsilon}{2}\right]^2 = \omega^4 \left[\frac{\mu^2\epsilon^2}{4} \left(1 + \frac{\mu^2\sigma^2}{4\omega^2\mu^2\epsilon^2} \times \frac{4}{\mu^2\epsilon^2}\right) \right]$$

$$\left(\beta^2 - \frac{\omega^2\mu\epsilon}{2}\right)^2 = \omega^4 \left[\frac{\mu^2\epsilon^2}{4} \left(1 + \frac{\sigma^2}{\omega^2\epsilon^2}\right) \right]$$

$$\beta^2 - \frac{\omega^2\mu\epsilon}{2} = \omega^2 \left[\sqrt{\frac{\mu^2\epsilon^2}{4} \left(1 + \frac{\sigma^2}{\omega^2\epsilon^2}\right)} \right]$$

$$\beta^2 = \frac{\omega^2\mu\epsilon}{2} + \frac{\omega^2\mu\epsilon}{2} \sqrt{1 + \frac{\sigma^2}{\omega^2\epsilon^2}}$$

$$\beta = \frac{\omega^2\mu\epsilon}{2} \left[1 + \sqrt{1 + \frac{\sigma^2}{\omega^2\epsilon^2}} \right]$$

$$\beta = \omega \sqrt{\frac{\mu\epsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega\epsilon}\right)^2} + 1 \right]}$$

$$v_p = f\lambda = \frac{\omega}{\beta} = \frac{1}{\sqrt{\frac{\mu\epsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega\epsilon}\right)^2} + 1 \right]}}$$

$$\text{and } d = \frac{2\pi}{\beta} = \frac{2\pi}{\omega \sqrt{\frac{\mu\epsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega\epsilon}\right)^2} + 1 \right]}}$$

$$\vec{E}(z,t) = \text{Real} [\vec{E}_0 e^{-\gamma z} e^{j\omega t}]$$

$$= e^{-\alpha z} \text{Real} [E_0 (\cos(\omega t - \beta z) + j \sin(\omega t - \beta z))]$$

Refractive index $n = \frac{c}{v_p} =$

A wave propagating in conducting medium is attenuated by a factor $e^{-\alpha z}$

Dielectric and Conductor

In EM field theory materials are classified as conductors and dielectrics but the boundary defining the two is not that sharp

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

$$= \vec{J}_c + \vec{J}_d$$

where $\vec{J}_c = \sigma \vec{E} \rightarrow$ conduction current density
~~Displacement~~ $\vec{J}_d = j\omega \epsilon \vec{E} \rightarrow$ displacement current density

$\frac{J_c}{J_d} = \frac{\sigma}{j\omega \epsilon} \rightarrow$ Mark the line b/w conductors and dielectrics
 $\rightarrow$ dissipation factor of dielectrics
 $\rightarrow$ loss tangent

1) $\frac{\sigma}{\omega \epsilon} \ll 1$ or $\frac{\sigma}{\omega \epsilon} < \frac{1}{100}$ - good dielectric

2) $\frac{\sigma}{\omega \epsilon} = 1$ $\frac{1}{100} < \frac{\sigma}{\omega \epsilon} < 100$ - quasi-dielectric

3) $\frac{\sigma}{\omega \epsilon} \gg 1$ $100 < \frac{\sigma}{\omega \epsilon}$ - good conductor

If $\sigma = 0 \rightarrow$ Perfect conductor.

Que:- A plane EM wave propagating through a medium with $\epsilon_r = 8$, $\mu_r = 2$ has $E = 0.5 e^{-\alpha z} \sin(10^8 t - \beta z) \hat{a}_x$ V/m. Determine (a) β (b) Loss tangent (c) Intrinsic impedance (d) wave velocity (e) H field, (2019) (10M)

Sol:- $\alpha = \omega \sqrt{\frac{\mu \epsilon}{2} \sqrt{1 + \left(\frac{\sigma}{\omega \epsilon}\right)^2} - 1} = \omega \sqrt{\frac{\mu_0 \epsilon_0 \mu_r \epsilon_r}{2} \sqrt{1 + \left(\frac{\sigma}{\omega \epsilon}\right)^2} - 1} = \frac{\omega}{c} \sqrt{\mu_r \epsilon_r \sqrt{1 + \left(\frac{\sigma}{\omega \epsilon}\right)^2} - 1} \because c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$

Let $x_0 = \sqrt{1 + \left(\frac{\sigma}{\omega \epsilon}\right)^2}$

$\alpha = \frac{\omega}{c} \sqrt{\frac{8 \times 2}{2} \sqrt{x_0 - 1}} = \frac{\omega}{c} \sqrt{8 \sqrt{x_0 - 1}} \rightarrow \sqrt{x_0 - 1} = \frac{\alpha c}{\omega \sqrt{8}} \quad \sqrt{x_0 - 1} = \frac{1}{\sqrt{8}} \left[x_0 = \frac{9}{8} \right]$

① $x_0 = \frac{9}{8} = \sqrt{1 + \left(\frac{\sigma}{\omega \epsilon}\right)^2} \quad \frac{81}{64} = 1 + \frac{\sigma}{\omega \epsilon} \quad \boxed{\frac{\sigma}{\omega \epsilon} = 0.5154} \rightarrow \text{loss tangent}$

② $\frac{\beta}{\alpha} = \sqrt{\frac{x_0 + 1}{x_0 - 1}} = \sqrt{\frac{9/8 + 1}{9/8 - 1}} = \frac{\sqrt{17}}{1} = \sqrt{17}$

$\beta = \alpha \sqrt{17} = \frac{\sqrt{17}}{3} = 1.374 \text{ rad/m} \quad \boxed{\beta = 1.374}$

③ $\eta = \frac{\sqrt{\mu/\epsilon}}{\sqrt{1 + \left(\frac{\sigma}{\omega \epsilon}\right)^2}} = \frac{120\pi \sqrt{2/8}}{\sqrt{9/8}} = 177.72$

$\tan 2\theta = \frac{\sigma}{\omega \epsilon} = 0.5154$

$\boxed{\theta = 13.63}$

④ $v_p = \frac{\omega}{\beta} = \frac{10^8}{1.374} = 7.278 \times 10^7 \text{ m/s}$

⑤ $H = \frac{E_0}{\eta} e^{-\alpha z} \sin(\omega t - \beta z - \theta) \hat{a}_y$
 $= \frac{0.5}{177.5} e^{-\alpha z} \sin(10^8 t - 1.374 z - 13.63^\circ) \hat{a}_y \text{ mA/m}$

Phase Velocity^U: It is the rate at which the phase of the wave propagates in space. This is the velocity at which the phase of any one frequency component of the wave travels. The phase velocity is given by λ and time period T as $v_p = \frac{\lambda}{T} \rightarrow \text{wavelength}$
or $v_p = \frac{\omega}{k} \rightarrow \text{angular frequency}$
 $\quad \quad \quad \rightarrow \text{wavenumber}$

Consider a basic sine wave $A \cos(kx - \omega t)$. After time t , the source has produced $\omega t / 2\pi = ft$ oscillations. After the same time, the initial wave front has propagated away from the source through space to

distance x to fit same no. of oscillation, $kx = \omega t$. So $v = x/t = \omega/k$.

Group Velocity: It is the velocity with which the overall envelope (shape) of the wave's amplitude (modulation) propagates through the space.

e.g. If a stone is thrown into a still pond, a circular pattern of waves with a quiescent appears. These expanding ring of wave is known as group wave within which one can discern individual wavelets of different wavelength traveling at different speed. The shorter waves travel faster than the group as a whole, but their amplitudes diminish as they approach the leading edge of the group. The longer waves travel more slowly, and their amplitudes diminish as they emerge from the trailing boundary of the group.

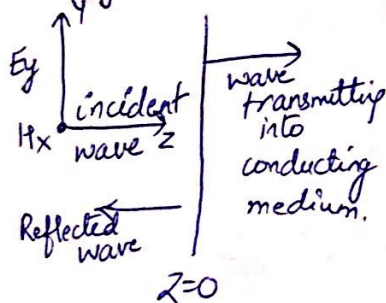
$$v_g = \frac{\omega_2 - \omega_1}{k_2 - k_1} = \frac{\Delta \omega}{\Delta k} = \frac{d\omega}{dk}$$



Depth of Penetration

It is a measure of how deep light or any EM radiation can penetrate into a material. In good conductors at RF the rate of attenuation is very rapid and wave may penetrate only a very short distance before being reduced to a negligibly small % of its initial strength.

- The depth of penetration may be defined as the depth in which the wave has been attenuated by amount $(1/e)$ or 37% approximately of its initial value. It is denoted by δ .
- Acc. to Beer-Lambert's law, the intensity of an EM wave inside a material falls off exponentially from the surface as $I(z) = I_0 e^{-\alpha z}$ where $\delta_p = \frac{1}{\alpha}$ denotes the depth of penetration.
- As the amplitude decreases exponentially by a factor $e^{-\alpha z}$ so obviously at a distance z which makes $\alpha z = 1$ the amplitude is only $(\frac{1}{e})$ times the original value at $z=0$.



Assume $z=0$ as the boundary of the conducting medium, z increases positively into conducting medium.

Equation of propagation becomes

$$E_y = E_0 e^{-(z/\delta)} e^{j(z/\delta)} \quad \text{where } \delta = \sqrt{\frac{2}{\omega \mu \sigma}} = \frac{1}{\sqrt{\omega \mu \sigma}}$$

$\vec{E}_y = \vec{E}_0 =$ maximum amplitude of the wave at the surface of the conducting medium.

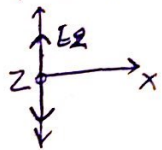
$$|\vec{E}_y| = \vec{E}_0 e^{-z/\delta} = \vec{E}_0 e^{-1} = \frac{\vec{E}_0}{e} \quad \text{at } z = \delta$$

$$E_y = \frac{1}{2.7} E_0 = 0.368 = 37\% E_0$$

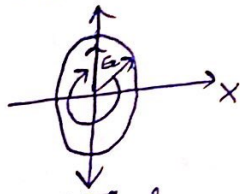
Thus when wave penetrates a distance δ , the field E_y decreases to 37%.

Electromagnetic Wave Polarization (2019)

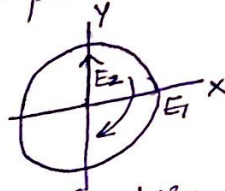
The polarization of a uniform plane wave refer to the time varying behaviour of the $\vec{E}$ at some fixed point in space



Linear polarization



Elliptical



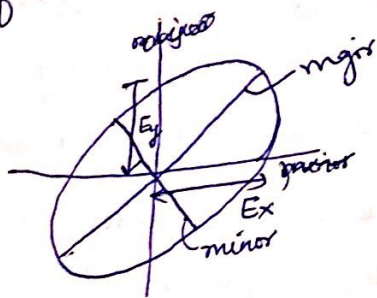
Circular

In linear polarization, the electrical vector is under all times remains only in one direction e.g. in y-direction. This wave is said to be linearly polarized in y-direction.

$$E_y = E_2 \sin(\omega t - \beta z)$$

$\vec{E}$ varies b/w +ve and -ve value of E_2 and the direction, confined to y direction only.

If the wave contains x and y components as well $\rightarrow$ elliptically polarised. At a fixed value of z, the electric field vector $\vec{E}$ rotates as a function of time.



The elliptical polarized wave is considered as a result of two plane (or linearly) polarized waves of same frequency. Both waves travel in +ve z direction but the plane of polarization of one wave is along x-axis while that of other is along y-axis.

Let E_x & E_y be $\vec{E}$ field of horizontally & vertically polarized wave respectively.

$$E_x = E_1 \sin(\omega t - \beta z)$$

$$E_y = E_2 \sin(\omega t - \beta z + \phi)$$

$\phi \rightarrow$ phase difference b/w E_x & E_y

$$\text{Now } \vec{E} = E_x \hat{a}_x + E_y \hat{a}_y$$

$$\text{At } z=0 \quad E_x = E_1 \sin \omega t \quad \rightarrow \quad \sin \omega t = \frac{E_x}{E_1}$$

$$E_y = E_2 \sin(\omega t + \phi)$$

$$E_y = E_2 [\sin \omega t \cos \phi + \sin \phi \cos \omega t]$$

$$E_y = E_2 \left[\frac{E_x}{E_1} \cos \phi + \sin \phi \sqrt{1 - \frac{E_x^2}{E_1^2}} \right]$$

$$\frac{E_y}{E_2} - \frac{E_x}{E_1} \cos \phi = \sin \phi \sqrt{1 - \frac{E_x^2}{E_1^2}}$$

$$\cos \omega t = \sqrt{1 - \frac{E_x^2}{E_1^2}}$$

square

$$\left(\frac{E_y}{E_2} \right)^2 + \left(\frac{E_x}{E_1} \right)^2 \cos^2 \phi - 2 \frac{E_y E_x}{E_2 E_1} \cos \phi = \sin^2 \phi - \frac{E_x^2}{E_1^2} \sin^2 \phi$$

$$= \left(\frac{E_y}{E_2} \right)^2 - \frac{2E_x E_y \cos \phi}{E_1 E_2} + \left(\frac{E_x}{E_1} \right)^2 = \sin^2 \phi$$

$$= \frac{E_y^2}{\sin^2 \phi E_2^2} - \frac{2E_x E_y \cos \phi}{E_1 E_2 \sin^2 \phi} + \frac{E_x^2}{E_1^2 \sin^2 \phi} = 1$$

$$[aE_y^2 - bE_x E_y + cE_x^2 = 1] \rightarrow \text{eqn of ellipse.}$$

Case I If $\phi = 180^\circ$ or $\phi = \pi$

$$\left(\frac{E_y}{E_2} \right)^2 \pm \frac{2E_x E_y}{E_1 E_2} + \left(\frac{E_x}{E_1} \right)^2 = 0 \quad \sin 0^\circ \sin 180^\circ = 0$$

$$\left(\frac{E_y}{E_2} \pm \frac{E_x}{E_1} \right)^2 = 0 \quad E_y = \pm \frac{E_2}{E_1} E_x \Rightarrow E_y = m E_x \rightarrow \text{eqn of line}$$

$\rightarrow$ The resultant wave is a plane (or linearly) polarized wave.

If $E_1 = 0 \rightarrow$ polarized in Y-direction

$E_2 = 0 \rightarrow$ polarized in X-direction

$E_1 = E_2$ and $\phi = 0 \rightarrow$ polarized at 45° (linearly polarised)

Case II $E_1 = E_2$ $\phi = \pm 90^\circ$

$$\frac{E_x^2}{E_1^2} - \frac{2E_x E_y \cos 90^\circ}{E_1 E_2} + \frac{E_y^2}{E_2^2} = \sin^2 90^\circ$$

$$\frac{E_x^2}{E_1^2} + \frac{E_y^2}{E_2^2} = 1 \rightarrow \text{eqn of circle.}$$

If $\phi = +90^\circ$ LHCP
 $\phi = -90^\circ$ RHCP

Poynting Theorem (ET 2019, 6M)

$\rightarrow$ It states that the vector product of $\vec{E}$ and $\vec{H}$ at any point is a measure of rate of energy flow per unit area at that point. The direction of power flow is perpendicular to $\vec{E}$ and $\vec{H}$

$$\vec{P} \equiv \vec{E} \times \vec{H}$$

$\vec{P}$ = Poynting radiation vector.

Proof: $\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$

$$\vec{J} = \nabla \times \vec{H} - \frac{\partial \vec{D}}{\partial t}$$

Multiplying by $\vec{E}$

$$\vec{E} \cdot \vec{J} = \vec{E} \cdot (\nabla \times \vec{H}) - \vec{E} \cdot \left(\frac{\partial \vec{D}}{\partial t} \right)$$

$$\vec{E} \cdot \vec{J} = \vec{H} \cdot (\nabla \times \vec{E}) - \nabla \cdot (\vec{E} \times \vec{H}) - \epsilon \vec{E} \cdot \frac{\partial \vec{E}}{\partial t}$$

$$\vec{E} \cdot \vec{J} = \vec{H} \cdot \left(-\frac{\partial \vec{B}}{\partial t} \right) - \nabla \cdot (\vec{E} \times \vec{H}) - \epsilon \vec{E} \cdot \frac{\partial \vec{E}}{\partial t}$$

$$\vec{E} \cdot \vec{J} = -\mu \vec{H} \cdot \frac{\partial \vec{H}}{\partial t} - \epsilon \vec{E} \cdot \frac{\partial \vec{E}}{\partial t} - \nabla \cdot (\vec{E} \times \vec{H})$$

Vector Identity

$$\nabla \cdot (\vec{E} \times \vec{H}) = \vec{H} \cdot (\nabla \times \vec{E}) - \vec{E} \cdot (\nabla \times \vec{H})$$

$$\vec{E} \cdot (\nabla \times \vec{H}) = \vec{H} \cdot (\nabla \times \vec{E}) - \nabla \cdot (\vec{E} \times \vec{H})$$

$$\mu H \cdot \frac{\partial H}{\partial t} = \frac{\mu}{2} \frac{\partial H^2}{\partial t} = \frac{\partial}{\partial t} \left(\frac{\mu H^2}{2} \right) \quad \text{Similarly } \epsilon \vec{E} \frac{\partial \vec{E}}{\partial t} = \frac{\partial}{\partial t} \left(\frac{\epsilon E^2}{2} \right)$$

$$\therefore \vec{E} \cdot \vec{J} = -\frac{\partial}{\partial t} \left(\frac{\mu H^2}{2} \right) - \frac{\partial}{\partial t} \left(\frac{\epsilon E^2}{2} \right) - \nabla \cdot (\vec{E} \times \vec{H})$$

$$= -\frac{\partial}{\partial t} \left[\frac{\mu H^2}{2} + \frac{\epsilon E^2}{2} \right] - \nabla \cdot (\vec{E} \times \vec{H})$$

Power per unit volume = $\frac{VI}{A} = \frac{VJ}{A} = EJ = \vec{E} \cdot \vec{J}$

Integrating throughout the volume

$$\int_V \vec{E} \cdot \vec{J} dV = -\frac{\partial}{\partial t} \int_V \left[\frac{\mu H^2}{2} + \frac{\epsilon E^2}{2} \right] dV - \int_V \nabla \cdot (\vec{E} \times \vec{H}) dV$$

↓ apply Divergence theorem

$$\oint_S (\vec{E} \times \vec{H}) \cdot d\vec{s}$$

$$-\oint_S \vec{E} \times \vec{H} \cdot d\vec{s} = \int_V \vec{E} \cdot \vec{J} dV + \frac{\partial}{\partial t} \int_V \left[\frac{\mu H^2}{2} + \frac{\epsilon E^2}{2} \right] dV$$

$\frac{1}{2} \epsilon E^2 \rightarrow$ stored $\vec{E}$ energy per unit volume
 $\frac{1}{2} \mu H^2 \rightarrow$ stored $\vec{H}$ energy per unit volume

In going power flux over surface S

Total power dissipated or generated within volume V

Rate of increase of total electromagnetic energy within the volume V enclosed by the surface at any instance.

If $\oint_S \vec{E} \times \vec{H} \cdot d\vec{s} \rightarrow$ ingoing or instantaneous power flux, then the $\oint_S \vec{E} \times \vec{H}$ represents total power flowing out of the volume.

Hence, $\vec{E} \times \vec{H} \equiv \vec{P}$ is instantaneous power density in Watts/m²

Poynting Vector: $\vec{P} \equiv \vec{E} \times \vec{H}$

$\vec{P}$ is known as Poynting vector. It is a measure of rate of flow of energy and its direction is the direction of power flow and $\perp$ to the plane containing $\vec{E}$ and $\vec{H}$.
 For plane wave, the direction of energy flow is the direction of propagation. So, Poynting vector offers a useful coordinate free way of specifying the direction of propagation.

Instantaneous & Average Poynting Vector

$$P_{inst} = V_{inst} \cdot I_{inst}$$

where $V_{inst} = \text{Re}[V e^{j\omega t}] = \text{Re}[V e^{j\omega_v} e^{j\omega_i t}] = |V| \cos(\omega t + \theta_v)$

$$\therefore V = |V| e^{j\theta_v}$$

Similarly $I_{inst} = |I| \cos(\omega t + \theta_i)$

Let $|V| = V_0$ and $|I| = I_0$

$$P_{inst} = \frac{1}{2} V_0 I_0 [\cos(\omega t + \theta_v) \cos(\omega t + \theta_i)]$$

$$= \frac{V_0 I_0}{2} [\cos(\theta_v - \theta_i) + \cos(2\omega t + \theta_v + \theta_i)]$$

Let $\theta = \theta_v - \theta_i$

$$P_{inst} = \frac{V_0 I_0}{2} [\cos \theta + \cos(2\omega t + \theta_v + \theta_i)]$$

average power

oscillating power

$P_{avg} = \frac{V_0 I_0}{2} \cos \theta \leftarrow$ inphase component of VI product
 $P_{reactance} = \frac{V_0 I_0}{2} \sin \theta \rightarrow$ out of phase component of VI product.

The presence of reactive component indicates that instantaneous power changes sign (reverse direction of flow) over ac cycle.

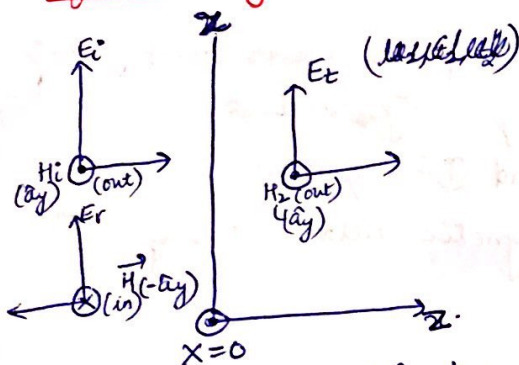
Complex Power: $P = \frac{1}{2} V I^*$

Illy $\boxed{\vec{P} = \frac{1}{2} \vec{E} \times \vec{H}^*}$

Reflection of a Plane EM Wave

When a plane wave propagates from one medium to another, it is partially reflected and partially transmitted.

Reflection by Perfect Dielectric (2017, 2018)



Y-direction is pointed out
Boundary at x
Direction of prop $\rightarrow z$

$\eta_1 = \sqrt{\frac{\mu_1}{\epsilon_1}}$, $\eta_2 = \sqrt{\frac{\mu_2}{\epsilon_2}} \rightarrow$ intrinsic impedance of mediums

$\eta_i = \frac{E_i}{H_i} \rightarrow \begin{cases} E_i = \eta_1 H_i \\ E_r = -\eta_1 H_r \\ E_t = \eta_2 H_t \end{cases}$

Acc. to eqn of continuity

$H_i + H_r = H_t$

$\frac{E_i}{\eta_1} + \frac{E_r}{-\eta_1} = \frac{E_t}{\eta_2}$

$\frac{1}{\eta_1} (E_i - E_r) = \frac{1}{\eta_2} E_t$

$\frac{1}{\eta_1} (E_i + E_r) = \frac{1}{\eta_2} (E_i + E_r)$

$E_i \left(\frac{1}{\eta_1} - \frac{1}{\eta_2} \right) = E_r \left(\frac{1}{\eta_2} + \frac{1}{\eta_1} \right)$

$E_i \left(\frac{\eta_2 - \eta_1}{\eta_2 \eta_1} \right) = E_r \left(\frac{\eta_1 + \eta_2}{\eta_1 \eta_2} \right)$ Reflection Coefficient

$\frac{E_r}{E_i} = \frac{\eta_1 + \eta_2}{\eta_2 - \eta_1}$ or $\frac{E_r}{E_i} = \frac{\eta_2 - \eta_1}{\eta_1 + \eta_2} = R$

$E_i + E_r = E_t$ and $H_i + H_r = H_t$

$E_i + E_r = E_t$

$\div$ by E_i

$1 + \frac{E_r}{E_i} = \frac{E_t}{E_i}$

$\frac{E_r}{E_i} = 1 + \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$

$\boxed{\frac{E_t}{E_i} = \frac{2\eta_2}{\eta_2 + \eta_1}}$

In term of H
 $\frac{H_r}{H_i} = \frac{-E_r}{E_i} = -\left(\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \right) = \frac{\eta_1 - \eta_2}{\eta_2 + \eta_1} \leftarrow RC$

$\frac{H_t}{H_i} = \frac{E_t}{\eta_2} \cdot \frac{1}{E_i/\eta_1} = \frac{2\eta_2 \cdot \eta_1}{\eta_1 + \eta_2} = \frac{2\eta_1}{\eta_1 + \eta_2} \leftarrow TC$

$\frac{H_t}{H_i} = \frac{\eta_1 - \eta_2}{\eta_1 + \eta_2}$, $\frac{H_t}{H_i} = \frac{2\eta_1}{\eta_1 + \eta_2}$

If $\mu_1 = \mu_2 = \mu_0$

For E
① $R = \frac{E_r}{E_i} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{\sqrt{\frac{\mu_0}{\epsilon_2}} - \sqrt{\frac{\mu_0}{\epsilon_1}}}{\sqrt{\frac{\mu_0}{\epsilon_2}} + \sqrt{\frac{\mu_0}{\epsilon_1}}} = \frac{\sqrt{\frac{1}{\epsilon_2}} - \sqrt{\frac{1}{\epsilon_1}}}{\sqrt{\frac{1}{\epsilon_2}} + \sqrt{\frac{1}{\epsilon_1}}} = \frac{\sqrt{\epsilon_1} - \sqrt{\epsilon_2}}{\sqrt{\epsilon_1} + \sqrt{\epsilon_2}}$

$T = \frac{2\sqrt{\frac{\mu_0}{\epsilon_2}}}{\sqrt{\frac{\mu_0}{\epsilon_2}} + \sqrt{\frac{\mu_0}{\epsilon_1}}} = \frac{2\sqrt{\epsilon_1}}{\sqrt{\epsilon_1} + \sqrt{\epsilon_2}}$

② for magnetic field

$R = \frac{\sqrt{\epsilon_2} - \sqrt{\epsilon_1}}{\sqrt{\epsilon_2} + \sqrt{\epsilon_1}}$

$T = \frac{2\sqrt{\epsilon_2}}{\sqrt{\epsilon_1} + \sqrt{\epsilon_2}}$

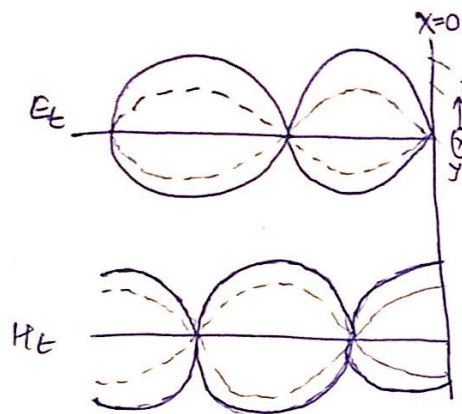
$E_i + E_r = E_t$
Divide by E_i

$1 + \frac{E_r}{E_i} = \frac{E_t}{E_i}$

$\frac{E_t}{E_i} = 1 + \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{2\eta_2}{\eta_2 + \eta_1}$

$\boxed{\frac{E_t}{E_i} = \frac{2\eta_2}{\eta_2 + \eta_1}} \leftarrow$ Transmission coefficient T

Case 1: Normal Incidence / Reflection by Perfect Conductor



Since no energy is absorbed. Hence the amplitude of $\vec{E}$ and $\vec{H}$ in reflected wave are same as in incident wave

Incident $\rightarrow E_i e^{-j\beta x}$
Reflected $\rightarrow E_r e^{+j\beta x}$ surface of conductor at $x=0$

Acc. to boundary condition $E_{\text{tangential}}$ is continuous at boundary

$$E_{\text{tan}} = E_i + E_r = 0$$

$$E_i = -E_r$$

$$E_t(x) = E_i e^{-j\beta x} + E_r e^{+j\beta x}$$

$$= E_i e^{-j\beta x} - E_i e^{+j\beta x} = -E_i \left(\frac{e^{j\beta x} - e^{-j\beta x}}{2j} \right) \cdot 2j$$

$$E_t(x) = -j(2E_i) \sin \beta x \quad \leftarrow \text{magnitude double \& } \sin \beta x \text{ is standing}$$

$$E_t(x, t) = -j(2E_i) \sin \beta x e^{j\omega t}$$

$$E_t(x, t) = \text{Re} [-j2E_i \sin \beta x e^{j\omega t}] \quad \text{--- (1)}$$

$$E_t(x, t) = 2E_i \sin \beta x \cdot \sin \omega t$$

The incident & reflected wave comb to produce a standing wave which does not propagate.

The magnitude of $\vec{E}$ varies sinusoidally with distance from reflecting plane. $\vec{E}$ is zero at surface and at $n\lambda/2$ (multiples of $\lambda/2$). Maximum values at $n\lambda/4$

$\vec{E}$ is reversed in phase on reflection in order to produce zero resultant field at the surface. So, $\vec{H}$ must be reflected without phase reversal

$$H_i = H_r$$

$$H_t(x) = H_i e^{-j\beta x} + H_r e^{+j\beta x} = 2H_i \frac{e^{j\beta x} + e^{-j\beta x}}{2} = 2H_i \cos \beta x \quad \text{--- (2)}$$

$$H_t(x, t) = \text{Re} [2H_i \cos \beta x e^{j\omega t}] = 2H_i \cos \beta x \cos \omega t \rightarrow \text{standing wave}$$

Maxima $\rightarrow$ surface + $n\lambda/2$

Minima $\rightarrow$ (zero points) $\rightarrow n\lambda/4$

for the boundary condition of H_t

Acc. to eqn (1) & (2) E_t and H_t are 90° out of phase because of j . Replacing $(-j)$ with $e^{-j(\pi/2)}$ and combining with $e^{j\omega t}$ we have $e^{j(\omega t - \pi/2)}$

$$\therefore E_t(x, t) = \text{Re} [e^{-j(\pi/2)} \cdot 2E_i \sin \beta x e^{j\omega t}]$$

$$= \text{Re} [2E_i \sin \beta x e^{j(\omega t - \pi/2)}]$$

$$= 2E_i \sin \beta x \cos(\omega t - \pi/2) \quad \text{--- (3)}$$

Comparing (2) & (3) we can see that there is a time difference of $\pi/2$ radians of E_t and H_t .

Que:- The electric field intensity of a uniform plane wave in air is 7500 V/m in the $\hat{y}$ direction. The wave is propagating in $\hat{x}$ direction at a frequency of 2×10^9 radians/sec. Find (1) frequency (2) wave length (3) time period (4) Amplitude of H (2018, 6.5M)

Soln:- $E_y = 7500 \cos(2 \times 10^9 t - \beta x)$

(1) wavelength = $\frac{v}{f} = \frac{3 \times 10^8}{2 \times 10^9 / 2\pi} = \frac{3\pi}{10} = \frac{3 \times 3.14}{10} = 0.943 \text{ m}$

(2) frequency = $\frac{\omega}{2\pi} = \frac{2 \times 10^9}{2\pi} = 318.3 \text{ MHz} = 3.18 \times 10^8 \text{ m/s}$

(3) Time = $\frac{1}{f} = \frac{1}{318.3} = 3.142 \text{ ns}$

(4) $\frac{E}{H} = \eta = 377 \Omega$ $H = \frac{E}{377} = \frac{7500}{377} = 19.9 \text{ A/m}$

$H = 19.9 \cos(2 \times 10^9 t - \beta x)$

Que:- The electric field intensity of a uniform plane wave propagating in a perfect dielectric medium having unity relative permeability is $E = 10 \cos(6\pi \times 10^7 t - 0.4\pi z) \hat{x} \text{ V/m}$. Find (1) Frequency (2) Wavelength (3) Phase velocity (4) Permittivity ϵ (5) Magnetic field intensity (2018, 6.5)

Soln:- $E = 10 \cos(6\pi \times 10^7 t - 0.4\pi z) \hat{x}$
 $\omega = 6\pi \times 10^7$

(1) frequency = $\frac{\omega}{2\pi} = \frac{6\pi \times 10^7}{2\pi} = 3 \times 10^7 \text{ Hz}$

(2) $\lambda = \frac{c}{f} = \frac{3 \times 10^8}{3 \times 10^7} = 10 \text{ m}$

(3) $V_p = \frac{\omega}{\beta} = \frac{6\pi \times 10^7}{0.4\pi} = \frac{60}{4} \times 10^7 = 1.5 \times 10^8 \text{ m/s}$

(4) $V_p = \frac{c}{\sqrt{\epsilon_r}} = 1.5 \times 10^8 = \frac{3 \times 10^8}{\sqrt{\epsilon_r}} \Rightarrow \sqrt{\epsilon_r} = 2 \Rightarrow \epsilon_r = 4$ $\eta = \sqrt{\frac{\mu}{\epsilon}} = \sqrt{\frac{1}{4}} = \frac{1}{2}$

(5) $B = \mu H = \frac{E}{\eta} = \frac{E_0 \cos(\omega t - \beta z) \hat{x}}{\eta} \text{ V/m} = \frac{10 \cos(6\pi \times 10^7 t - 0.4\pi z)}{\frac{1}{2}} \hat{x}$
 $B = 20 \cos(6\pi \times 10^7 t - 0.4\pi z) \hat{x}$

Que:- Calculate skin depth of propagation constant γ and wave velocity v at a frequency of 1.6 MHz in Aluminium whose $\sigma = 38.2 \text{ Ms}$ and $\mu_r = 1$. (2017)

Soln:- $\delta = \frac{1}{\alpha}$ where $\alpha = \sqrt{\frac{\omega \mu \sigma}{2}}$ $\omega = 2\pi f$

$\delta = 0.072 \times 10^{-3}$ $\alpha = \sqrt{\frac{1}{\pi f \mu \sigma}} = \sqrt{\frac{1}{3.14 \times 1.6 \times 10^6 \times 1 \times 38.2}}$